

# General Topology

Selected Topics on Mathematical Physics | Autumn 2026

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## I Prerequisites

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Students are expected to be familiar with the following topic:

- Set theory
- Elementary calculus

## II Course Structure

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The course meets twice per week for lectures. In addition, a weekly tutoring session is held to support work on the problem sheets; attendance is strongly encouraged, as the sheets form an integral part of the course.

|                  | Format                 | Duration   |
|------------------|------------------------|------------|
| Regular sessions | Theory and derivations | 80 minutes |
| Tutoring session | Problem sheet workshop | 90 minutes |

Problem sheets should be attempted *before* the tutoring session. Hints and solutions are discussed in the tutoring session and collected shortly thereafter.

## III Course Description

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This course provides a rigorous introduction to the foundational structures of general topology — the mathematical study of continuity, convergence, and spatial properties in their most abstract form. Starting from the first generalisations of  $\mathbb{R}^n$ , the metric spaces, the course develops the motivation behind the definition of a topology on a set via the open subset criterion for continuity, it goes through the amazing properties of these general spaces, culminating in the deep classical theorems that underpin large parts of modern mathematics, making emphasis on the particular cases of multivariable calculus. The material is essential preparation for graduate study in geometry, analysis, algebraic topology, and mathematical physics, and simultaneously serves as a training ground for abstract mathematical reasoning.

## IV Contents

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1. **Metric Spaces** — Euclidean spaces; the Euclidean norm; metrics; continuity; convergent and Cauchy sequences; open subset criterion for continuity.
2. **Topological Spaces** — Topologies; convergence and continuity; Hausdorff spaces; bases and countability; topological manifolds.

3. **New Spaces from Old** — Subspaces; product spaces; quotient spaces; adjunction spaces; topological groups and group actions.
4. **Connectedness and Compactness** — Disconnectedness; open covers and subcovers; compactness; local compactness; paracompactness; proper maps.

## V References

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*The star rating reflects the mathematical sophistication of each text: (★) accessible with the course prerequisites; (★★) requires additional mathematical maturity; (★★★) intended for readers revisiting the subject — not recommended if this is your first encounter with the material.*

- [1] Lee, J. M. *Introduction to Topological Manifolds*. 2nd ed. Springer, 2011.  
[Primary (★)]
- [2] Szekeres, P., *A Course in Modern Mathematical Physics: Groups, Hilbert Space and Differential Geometry*. Cambridge University Press, 2004.  
[Supplementary (★★)]